

Peradeniya Botanical Garden

Data Collection Guide

Hybrid Pollination & Compatibility Analysis

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R26-SE-018 | SLIIT

Date: August 18, 2026

MISSION SUMMARY

Collect orchid plant images and pollination compatibility data for two ML models:
Level 1: Single plant suitability assessment (need more unhealthy/moderate plant images)
Level 2: Two-parent compatibility prediction (need pair data with known outcomes)
MOST IMPORTANT: Talk to the Orchid House staff about breeding records and hybrid parents!

WHAT IS A HYBRID ORCHID?

A hybrid is a new plant created by cross-pollinating two different parent plants (usually different species or varieties). The breeder takes pollen from Plant A and places it on the flower of Plant B. If compatible, seeds form and grow into a new hybrid plant with features from both parents.

Example:

Parent A: *V. coerulea* (blue flowers)

Parent B: *V. sanderiana* (large pink flowers)

==> Hybrid: *V. Rothschildiana* (large blue-purple flowers)

The hybrid exists = proof that Parent A + Parent B are COMPATIBLE.

This is real training data for your Level 2 model!

What to look for at the garden:

Look for plant labels that show parentage, like:

"Vanda Rothschildiana (*V. coerulea* x *V. sanderiana*)"

This tells you: *coerulea* + *sanderiana* = Successful cross!

Photograph the label AND the hybrid plant.

WHY THIS MATTERS

Every hybrid you find = one confirmed 'Successful' data point for your dataset.

Every failed cross the staff mentions = one 'Failed' data point.

This is the HARDEST data to get, so prioritize conversations with staff!

LEVEL 1: INDIVIDUAL PLANT IMAGES

For the suitability model. Your dataset is imbalanced - you need MORE unhealthy plants.

Collection Targets:

Category	What to Collect
Unhealthy plants (NOT SUITABLE)	30-50 images [HIGHEST PRIORITY]
Moderate plants	30-50 images [HIGH PRIORITY]
Healthy plants (SUITABLE)	10-20 images (already have 230+)
Flower close-ups	20-30 images [MEDIUM PRIORITY]
Named species	3-5 different Vanda species

What UNHEALTHY plants look like:

- Yellow or brown leaves
- Few leaves, thin/damaged roots
- Spots, black marks, fungal patches on leaves
- Wilting or drooping
- Pale/washed-out leaf color

What MODERATE plants look like:

- Mix of green and slightly pale leaves
- Minor discoloration but generally alive
- Some leaves healthy, some showing stress

Camera Tips:

- Use HDR mode (keep consistent with existing data)
- Take 5-10 photos per plant from different angles
- Include the WHOLE plant (roots, stem, leaves)
- Avoid flash - use natural light
- Also take close-up shots of leaves and flowers
- Photograph the species label BEFORE the plant

LEVEL 2: COMPATIBILITY PAIR DATA

For the compatibility model. You need data about PAIRS of plants and whether their cross-pollination was successful or not.

How to collect pair data:

Step 1: Find hybrids in the Orchid House

- Look for labels showing hybrid parentage (e.g., 'V. X = V. A x V. B')
- Photograph: the label, the hybrid plant, and parent species if present
- Record: Parent A species + Parent B species = SUCCESSFUL

Step 2: Talk to the Orchid House staff

- Introduce yourself: SLIIT research project on Vanda hybrid pollination
- Ask: "Which Vanda crosses have been done here?"
- Ask: "Which crosses work well? Which usually fail?"
- Ask: "Are there breeding records I could see?"
- Ask: "Do cross-species Vanda pollinations usually succeed?"
- Record EVERYTHING they say - even casual comments are data!

Step 3: Photograph species pairs

- For every different Vanda species you find, photograph it
- Record the species name from the label
- Ask staff if any two species you found can be cross-pollinated

Vanda species to look for:

Species	Identifying Feature
V. tessellata	Native to Sri Lanka - very likely to be there!
V. coerulea	Blue/violet flowers - popular in gardens
V. teres	Cylindrical (pencil-like) leaves
V. tricolor	Multi-colored petals with spots
V. denisoniana	Yellow/white fragrant flowers
V. sanderiana	Large pink/white flowers
Any other Vanda	Note the exact name from the label

QUESTIONS TO ASK THE STAFF

INTRODUCTION SCRIPT

Say: "Hi, I am a final year IT student at SLIIT doing a research project on predicting Vanda orchid hybrid pollination compatibility using machine learning. I am collecting data about which Vanda crosses are successful and which fail. Could you help me with some information?"

Must-Ask Questions:

About Hybrids:

- ☐ Do you have any Vanda hybrid orchids here? What are their parent species?
- ☐ Which Vanda species crosses are most successful?
- ☐ Are there any crosses that were tried but always fail?
- ☐ Does cross-species pollination usually work for Vandas?

About Plant Health & Pollination:

- ☐ How do you select parent plants for pollination?
- ☐ Do you avoid pollinating diseased or weak plants?
- ☐ How important is leaf condition for pollination success?
- ☐ What makes a plant 'ready' for pollination?

About Records:

- ☐ Do you keep breeding/pollination records?
- ☐ Could I photograph any breeding charts or records?
- ☐ Are there any research papers from work done here?

About Outcomes:

- ☐ When a cross succeeds, how long until you see results?
- ☐ Do some successful crosses produce weaker flowers?
- ☐ What is the typical success rate for Vanda pollination?

INDIVIDUAL PLANT DATA SHEET

(Fill one per plant photographed. Use phone notes app if easier.)

Plant #1 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #2 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #3 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

INDIVIDUAL PLANT DATA SHEET

(Fill one per plant photographed. Use phone notes app if easier.)

Plant #4 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #5 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #6 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

INDIVIDUAL PLANT DATA SHEET

(Fill one per plant photographed. Use phone notes app if easier.)

Plant #7 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #8 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

Plant #9 (ID: PBG_____)

Species Name: (from label board)

Photos Taken: Count: _____

Leaf Condition: healthy / moderate / weak (circle one)

Leaf Count: few / medium / many (circle one)

Plant Strength: strong / moderate / weak (circle one)

Disease Visible: yes / no (circle one)

Flower Present: yes / no (circle one)

Flower Condition: good / moderate / weak / N/A (circle one)

Suitability (your judgment): Suitable / Moderate / Not Suitable

Notes: _____

COMPATIBILITY PAIR DATA SHEET

(Fill one per pair. Record hybrid info OR staff knowledge about crosses.)

Pair #1 (ID: PAIR_____)

PARENT A:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

PARENT B:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

CROSS INFORMATION:

Cross Type:

same-species / cross-species (circle one)

Known Outcome:

successful / failed / partially / unknown

Data Source:

hybrid label / staff said / literature / unknown

Flower Quality:

high / medium / low / unknown

Notes:

Pair #2 (ID: PAIR_____)

PARENT A:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

PARENT B:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

CROSS INFORMATION:

Cross Type:

same-species / cross-species (circle one)

Known Outcome:

successful / failed / partially / unknown

Data Source:

hybrid label / staff said / literature / unknown

Flower Quality:

high / medium / low / unknown

Notes:

COMPATIBILITY PAIR DATA SHEET

(Fill one per pair. Record hybrid info OR staff knowledge about crosses.)

Pair #3 (ID: PAIR_____)

PARENT A:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

PARENT B:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

CROSS INFORMATION:

Cross Type:

same-species / cross-species (circle one)

Known Outcome:

successful / failed / partially / unknown

Data Source:

hybrid label / staff said / literature / unknown

Flower Quality:

high / medium / low / unknown

Notes:

Pair #4 (ID: PAIR_____)

PARENT A:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

PARENT B:

Species Name:

Plant Condition:

healthy / moderate / weak

Linked Plant ID:

(PBG_____ if you photographed it)

CROSS INFORMATION:

Cross Type:

same-species / cross-species (circle one)

Known Outcome:

successful / failed / partially / unknown

Data Source:

hybrid label / staff said / literature / unknown

Flower Quality:

high / medium / low / unknown

Notes:

QUICK CHECKLIST - BEFORE YOU LEAVE

Check off each item before leaving the garden to make sure you got everything:

Level 1 - Plant Images:

- ☐ Photographed unhealthy/diseased plants (target: 30-50 images)
- ☐ Photographed moderate-health plants (target: 30-50 images)
- ☐ Photographed healthy plants (target: 10-20 images)
- ☐ Took flower close-up photos (target: 20-30)
- ☐ Photographed at least 3 different named Vanda species
- ☐ Took a photo of each species label BEFORE the plant
- ☐ Filled in plant data sheets / phone notes for each plant

Level 2 - Compatibility Data:

- ☐ Found and photographed hybrid orchids with parent labels
- ☐ Talked to Orchid House staff about breeding
- ☐ Recorded staff answers about successful/failed crosses
- ☐ Recorded at least 5-10 plant pairs
- ☐ Photographed any breeding charts/info boards
- ☐ Filled in pair data sheets for each pair

General:

- ☐ All photos are in HDR mode with natural light
- ☐ Multiple angles per plant (5-10 photos each)
- ☐ Species names recorded accurately from labels
- ☐ Backed up photos to cloud before leaving

REMEMBER

The CONVERSATION with Orchid House staff is potentially MORE valuable than the photos!

Be polite, explain your research, and record everything they tell you.

Even casual comments like 'that cross never works' = valuable data for your model.

Good luck tomorrow!